

CASE STUDY MANUAL

PROMPT ENGINEERING FOR GENERATIVE AI

TITLE: AI-Assisted Literature Review, Research Gap Analysis and Publication-Oriented Review Paper Development using Prompt Engineering

COURSE

Prompt Engineering for Generative AI

FINAL OBJECTIVE

At the end of this case study, every student must produce:

CASE STUDY COMPLETION REQUIREMENTS

Students must complete and submit all of the following components as evidence of their research process, prompt engineering activities, literature review, and review paper development.

Component No.	Submission Component	Description
1	Research Paper Repository	Collection of minimum 30 research papers with bibliographic details
2	Literature Review Master Matrix	Completed analysis table for all selected papers
3	Prompt Engineering Portfolio	Complete record of prompts used, techniques applied, and outcomes obtained
4	Comparative Analysis Report	Domain-specific comparison of methodologies, datasets, tools, architectures, or evaluation metrics
5	Limitation Mining and Research Gap Report	Evidence-based identification of limitations and derivation of research gaps
6	Future Research Directions Report	Proposed future research opportunities and novel ideas
7	Publication-Oriented Review Paper	Final review/survey paper (4000–6000 words) in IEEE format
8	Originality Validation Report	AI detection, plagiarism, grammar, and readability analysis

Component No.	Submission Component	Description
9	Technical Presentation	Structured 15-slide presentation summarizing the entire case study

IMPORTANT REQUIREMENTS (FOR AI BASED RESEARCH REVIEW)

Research Papers

Minimum: **30 Research Papers**

Preferred: **40–50 Papers**

Publication Sources

Only from:

- IEEE
- Springer
- ACM
- Elsevier
- MDPI
- Wiley
- arXiv (Recent Trends Only)

Publication Years

2015–Present

Final Review Paper Requirements

Parameter	Requirement
Word Count	4000–6000
References	Minimum 30
Similarity Index	<10%
AI Detection	<10%

Parameter	Requirement
Grammar Score	>90%
Citations	IEEE Format

PHASE 1

RESEARCH DOMAIN IDENTIFICATION

Student Information

Field	Details
Student Name	
Enrollment No	
Department	
Batch	
Date	

Research Domain Selection

Choose one.

Domain	Selected
Generative AI	
Machine Learning	
Deep Learning	
Computer Vision	
Cybersecurity	
Cloud Computing	
IoT	

Domain	Selected
FPGA	
VLSI	
Embedded Systems	
Data Analytics	
Other	

Topic Narrowing Exercise

Example

AI

↓

Generative AI

↓

RAG

↓

Hybrid RAG

↓

Enterprise RAG Systems

Student Topic Definition

Level	Topic
Broad Domain	
Sub Domain	
Research Area	
Specific Focus	
Final Review Topic	

Descriptive Questions

Q1. Why did you select this topic?

Q2. What industrial relevance does this topic have?

Q3. Why is this topic currently important?

Q4. What challenges exist in this area?

PHASE 2

RESEARCH PAPER DISCOVERY

TASK

Collect and shortlist minimum 30 papers.

Paper Repository Master Sheet

This is the most important sheet.

Sr.	Paper Title	Author(s)	Year	Publisher	DOI/Link	Citation Count	Relevance (1–5)
1							
2							
3							

(Continue until 30+ papers)

Analysis Questions

Q1

How did you identify these papers?

Q2

What keywords did you use?

Q3

How did AI assist in paper discovery?

PHASE 3

LITERATURE REVIEW MASTER MATRIX

TASK

Read all 30 papers and populate the following table.

Paper ID	Research Problem	Methodology	Dataset/Benchmark	Evaluation Metrics	Key Findings	Advantages	Limitations	Future Scope
P1								
P2								
P3								

(30 Entries)

Reflection

Which methodology appears most frequently?

Which dataset appears most frequently?

Which evaluation metric appears most frequently?

PHASE 4

PROMPT ENGINEERING ACTIVITY LOG

TASK

Every AI interaction must be documented.

Prompt Activity Sheet

Activity	Objective	Prompt Used	Prompt Technique	Output Quality (1–5)	Improvements Made
Paper Summary					
Dataset Analysis					
Method Comparison					
Gap Analysis					

Reflection Questions

Why was this prompting technique selected?

Did AI hallucinate?

How was the output verified?

PHASE 5

RESEARCH EVOLUTION ANALYSIS

TASK

Create a technology timeline.

Technology Evolution Sheet

Year	Major Contribution	Paper ID	Impact
2015	CNN-based object detection introduced	P3	Improved detection accuracy compared to traditional machine learning approaches

Year	Major Contribution	Paper ID	Impact
2016			
2017			

Descriptive Questions

What technological shifts occurred over the last decade?

Which techniques became obsolete?

Which techniques became dominant?

PHASE 6

DOMAIN-SPECIFIC COMPARISON STUDY

TASK

Select an appropriate comparison matrix.

AI / ML

Paper	Model	Dataset	Accuracy	Precision	Recall	F1	Limitation
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Computer Vision

Paper	Technique	Dataset	mAP	Accuracy	Complexity	Limitation
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VLSI

Paper	Architecture	Area	Power	Delay	Throughput	Limitation
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FPGA

Paper	FPGA Device	Resource Utilization	Latency	Throughput	Limitation
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Cybersecurity

Paper	Technique	Dataset	Detection Rate	False Positive Rate	Limitation
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PHASE 7

LIMITATION MINING

TASK

Extract every limitation mentioned in all papers.

Limitation Repository

Paper ID	Limitation 1	Limitation 2	Limitation 3
P1			
P2			

Limitation Frequency Table

Limitation Category	Frequency
Scalability	
Security	
Dataset Issues	
Cost	
Generalization	

PHASE 8

RESEARCH GAP DISCOVERY

IMPORTANT

Students must not directly ask AI:

Find research gaps.

Instead derive gaps from limitations.

Research Gap Sheet

Gap ID	Supporting Papers	Evidence	Why It Matters
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Descriptive Questions

Why is this a genuine research gap?

Which papers support this gap?

What industrial problem can this solve?

PHASE 9

FUTURE RESEARCH DIRECTIONS

Future Scope Matrix

Gap	Proposed Research Direction	Expected Impact	Feasibility
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Novel Research Ideas

Minimum:

3 Original Ideas

For each idea answer:

Question	Response
Problem Solved	
Supporting Papers	
Why Novel?	
Expected Impact	

PHASE 10

REVIEW PAPER DRAFTING

TASK: Use AI as:

Research Analyst

Technical Reviewer

Academic Writer

Journal Reviewer

Section Development Tracker

Section	AI Role Used	Prompting Technique Used	Human Modifications Made	Final Status
Abstract	Research Writer	Role Prompting	Added domain insights and citations	Completed
Introduction	Research Analyst	Chain-of-Thought	Reorganized flow and references	Completed
Literature Review	Research Reviewer	Prompt Chaining	Added comparisons across papers	Completed
Comparative Analysis	Technical Analyst	ReAct	Added evaluation metrics	Completed
Research Gaps	Research Mentor	CoT + Role Prompt	Derived gaps manually from limitations	Completed
Future Scope	Research Consultant	Role Prompting	Added industrial relevance	Completed
Conclusion	Academic Writer	Prompt Chaining	Rewritten in own language	Completed

PHASE 11

HUMANIZATION & ORIGINALITY IMPROVEMENT

Humanization Log

Section	AI Draft Words	Final Words	Major Changes Made

Reflection Questions

What domain knowledge did you manually add?

What content was removed because it sounded AI-generated?

Which sections required major rewriting?

PHASE 12

AI DETECTION & PLAGIARISM VALIDATION

Validation Report

Parameter	Result
Similarity Index	
AI Detection Score	
Grammar Score	
Readability Score	

Mandatory Requirements

Metric	Required
Similarity	<10%
AI Detection	<10%
Grammar	>90%
Readability	Good

PHASE 13

FINAL REVIEW PAPER SUBMISSION

Final Paper Structure

1. Title
2. Abstract

3. Keywords
4. Introduction
5. Research Methodology
6. Literature Review
7. Comparative Analysis
8. Research Trends
9. Research Gaps
10. Future Research Directions
11. Conclusion
12. References

FACULTY EVALUATION RUBRIC (100 MARKS)

Criteria	Marks
Topic Selection & Scope	5
Paper Collection (30+)	10
Literature Review Master Matrix	15
Prompt Engineering Portfolio	10
Research Evolution Analysis	10
Comparative Study	10
Limitation Mining	10
Research Gap Identification	10
Future Scope & Novel Ideas	10
Humanization & Originality	5
Final Review Paper Quality	5
Total	100